Chapter 9 • Similar Figures

Solved Exercise 9.1

1. Determine whether the solids with dimensions (3,4,5) cm and (4.5,6,7.5) cm are similar.

Yes. $\frac{4.5}{3}=\frac{6}{4}=\frac{7.5}{5}=1.5$, so all corresponding lengths have the same scale factor.

2. Prove triangles with sides (6,9,12) cm and (10.5,15.75,21) cm are similar.

$\frac{10.5}{6}=1.75$, $\frac{15.75}{9}=1.75$ and $\frac{21}{12}=1.75$.

3. Given ΔABC~ΔDEF, AB=12, AC=20, BC=16 and DE=6 cm, find DF and EF.

Scale from ABC to DEF=DE/AB=6/12=1/2.

DF = $(\frac{1}{2})$AC

= 10 cm

EF = $(\frac{1}{2})$BC

= 8 cm.

DF=10 cm; EF=8 cm.

4. Find x in each similar-triangle diagram.

(i) $\frac{x}{1.2}=\frac{7.5}{3}$

⇒ x=3 cm.

(ii) EF∥CD gives BE/BC=BF/BD: 6/(6+x)=8/11 ⇒ x=2.25 cm.

(iii) Equal marked angles and vertical angles give $\frac{x}{2.1}=\frac{2.5}{2.4}$ ⇒ x=2.1875≈2.19 cm.

5. A stair is 20 cm wide and 16 cm deep/high; an 8 cm-high box of width x fits beneath the plank.

The small triangle above the box has height 16−8=8 cm and base x. By similarity, $\frac{x}{20}=\frac{8}{16}$

x=10 cm.

6. A 1.8 m man casts a 0.76 m shadow; a pole casts a 3 m shadow. Find its height.

At the same time, height/shadow is constant: h/3=1.8/0.76.

h=$\frac{5.4}{0.76}$≈7.11 m.

7. In a right triangle, the altitude to the hypotenuse divides it into segments 6 cm and x; the leg adjacent to the 6 cm segment is 10 cm. Find x, altitude y and other leg z.

Leg rule: $10^2=6(6+x)$

⇒ 100=36+6x

⇒ x=$\frac{32}{3}$=10⅔ cm.

Altitude rule: $y^2=6x=64$

⇒ y=8 cm.

Other leg rule: $z^2=x(6+x)=(32/3)(50/3)=\frac{1600}{9}$ ⇒ z=$\frac{40}{3}$=13⅓ cm.

x=10⅔ cm, y=8 cm, z=13⅓ cm.

8. In trapezoid ABCD, AB∥CD and diagonals meet at E. Prove ΔABE~ΔCDE; if AB=8, CD=4, AE=3 cm, find CE.

∠AEB=∠CED (vertically opposite). Alternate angles from AB∥CD give ∠EAB=∠ECD and ∠EBA=∠EDC; hence AA similarity.

AB/CD=AE/CE

⇒ $\frac{8}{4}=\frac{3}{CE}$

Similarity proved; CE=1.5 cm.

9. A regular dodecagon of perimeter 72 cm has each side multiplied by $\frac{1}{\sqrt{2}}$ Find the new side length.

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Original side$=\frac{72}{12}=6$ cm; scaled side$=\frac{6}{\sqrt{2}}$
$3\sqrt{2}\approx4.24$ cm.

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